

Inequalities of Polynomials

Key Points

To solve polynomial inequalities first set the equation against zero then solve this equation to find roots, sketch $g(x)$ and state the domain of x such that the inequality is satisfied

Remember the factor theorem often helps to find some factors

Remember the rules of inequalities and be careful multiplying by variables i.e. be careful when multiplying by $x, x^2, \dots$ or $f(x)$

Basic Skills

Q1. Solve the following quadratic equations

a) $4x^2 - 22x + 24 = 0$ b) $5x^2 + 6x - 12 = 0$

Q2. State the factor/remainder theorem

Q3. Given that $x - 2$ is a factor of $f(x) = 3x^3 + x^2 - 12x - 4$ find the roots of $f(x)$

Q4. Solve the following quadratic inequalities

a) $4x^2 - 22x + 24 \geq 0$

b) $2x^2 + 2x - 5 < 0$

c) $x^2 - 5 \geq 3x + 5$

Q5. Solve $0 \leq x^2 - 2x - 3 \leq 5$

Competency Skills

Q6. a) Sketch the graph of $y = (x + 3)(x - 2)(4x + 5)$

b) Hence solve the inequality $(x + 3)(x - 2)(4x + 5) > 0$

Q7. Solve the inequality $(x + 3)(x - 2)(4x + 5)(3 - x) \leq 0$

Q8. Solve the inequality $6x^4 - 5x^2 + 6x > 13x^3 - 6x + 4$

Q9. a) Show the inequality $x^2 - 7 \geq \frac{6}{x}$ can be written as $x^4 - 7x^2 - 6x \geq 0$

b) State what we will have introduced in part a)

c) Hence solve the inequality $x^2 - 7 \geq \frac{6}{x}$

d) Explain why we cannot say the inequality $x^2 - 7 \geq \frac{6}{x}$ is equivalent to $x^3 - 7x - 6 \geq 0$

Advanced Skills

Q8. a) Solve the equation $x^2 - x - 4 = x - x^2$

b) Hence solve the simultaneous inequalities $x^2 - x - 4 > x - x^2$ and $2x - 5 \leq 0$

Q9. a) Sketch the graphs of $y = x^5$ and $y = \frac{1}{x}$ labelling the points at which they intersect

b) Hence solve the inequality $x^5 \leq \frac{1}{x}$

c) Compare your answer to part b) to solving the inequality $x^6 \leq 1$ and $x^7 \leq x$ commenting on the difference

Q10. Solve the inequality $x + 2 < \frac{1}{x-2}$

Extension

Q11. Why might it be complicated to consider the inequality $\frac{x^2 + 5x - 6}{x(x-2)(2x+3)} > 1$

Hint: Consider problems with making it a polynomial inequality

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